

Chapter 3

Caland verbs in Celtic

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This study offers a comprehensive examination of Caland verbs in the Celtic languages, an area that has previously received little attention within the broader investigation of the Caland system across Indo-European languages. By identifying and analyzing twenty Celtic verbs associated with the Caland system, we determine that eleven likely represent inherited forms, while the remaining verbs appear to be inner-Celtic innovations. Through detailed philological, phonological, and morphological analysis, we assess the productivity of root-derived deadjectival verb formations in Proto-Celtic. Despite the relatively limited number of verbs and their distribution across various verbal stem types, the findings suggest that the Caland system retained a marginal but discernible degree of productivity in the verbal domain of the Celtic languages. This conclusion is further supported by the existence of at least two neo-roots in the nominal domain, demonstrating parallel processes of morphological innovation.

1 Introduction

When it comes to the systematic study of the Caland system and Caland morphology in the Indo-European languages, most of the relevant research so far has been conducted on the basis of Indo-Iranian, Greek, Hittite, and Latin (e.g., Nussbaum 1976; Rau 2009; Rau 2013; Lowe 2016, etc.), Balto-Slavic (e.g., Majer 2017), or has been focused on the reconstruction of the system in the proto-language (e.g., Bozzone 2016; on the history of scholarship see Dell’Oro 2015).

The Celtic languages have played a subordinate role in establishing the dynamics of the system. When Celtic material has been mentioned at all, its function has been ancillary rather than constitutional, serving as a means to provide evidence for an already established framework rather than being exploited to gain



new insights into the Caland complex as a whole. And while a couple of book chapters and conference talks on Celtic such as De Bernardo Stempel (1999: 529–537, 2003), Höfler (2016), and Kahl (2018) (on Gaulish) have in recent years been devoted to the investigation of the nominal domain, the Caland-associated verbs still lack a comprehensive treatment.¹

Very broadly speaking, Caland verbs are verbs that belong to Caland roots, or roots with a “Caland profile” (see Nussbaum 2022). These roots possess an adjectival meaning that typically falls into one of the categories known as ‘property concepts’ (i.e., dimensions, physical properties, colours, speed, age, value, human propensities, etc.), such as, for instance, ‘big’, ‘small’, ‘red’, ‘fast’, ‘slow’, ‘old’, ‘new’, ‘good’, ‘bad’, etc. (see Rau 2009: 78–79; Balles 2003: 18–20; Balles 2006: 269–287; see also the introduction to this volume, Sections 1.2 and 3.1). Despite their adjectival meaning (‘X’), these roots do not (or at least not normally) surface as root adjectives. If a root noun is attested, it typically has the meaning of an adjectival abstract (‘X-ness’) or a concrete noun (‘thing/person that is X’). Compare the root **b^herǵ^h-* ‘high’ with the adjective **b^herǵ^h-(é)nt-* ‘high’ (Gaulish, Old British ethnonym sg. *Brigans*, pl. *Brigantes*, Βριγαντες, fem. *Brigantiae* (dat.), OIr. *Brigit* < **b^herǵ^h-nt-ih₂-* (see also Stifter forthcoming); cf. Vedic *bṛhánt-*, *bṛhatí-* ‘big, high, noble’, Young Avestan *barāzant-* ‘high’) and the root noun **b^herǵ^h-* ‘height, high spot’ (Old Irish *brí*, gen.sg. *breg* f. ‘hill’; cf. Young Avestan *barāz-*, nom.sg. *bars* ‘height, mountain; high’, Proto-Germanic **burg-* ‘castle’ > Gothic *baurgs* f. ‘city, tower’, Old High German *burg* etc. ‘castle’), or the root **(h₁)reud^h-* ‘red’ with the adjective **(h₁)re/ouð^h-o-* ‘red’ (Old Irish *ród*, *rúad*, Welsh *rhudd*; cf. Latin *rūfus*, *rōbus*, Proto-Germanic **rauda-*, Lithuanian *raūdas*, Old Church Slavonic *rudŏ*) and the root noun **(h₁)rud^h-* ‘redness’ (Old Irish *rú*, dat.sg. *roid* f. ‘redness, red dye plant’). As with **b^herǵ^h-(é)nt-* ‘high’ and **(h₁)rouð^h-o-* ‘red’, the corresponding base adjective X usually exhibits derivational suffixes such as **-ro-*, **-u-*, **-o-*, **-nt-*, **-mo-*, **-no-*, a fact that in and of itself would not be overly surprising. However, all (seemingly) deadjectival derivatives are consistently root-derived, not stem-derived. This is true for typical deadjectival formations such as the comparative and superlative (e.g., **tép-nt-* ‘hot’ > Old Irish *tee*, *té*, nom.pl. *téit*, but comparative **tep-īās* > Old Irish *téo*, *teo*, *teou*), deadjectival abstracts (e.g., **treks-no-* ‘strong’ > Old Irish *trén*, but **treks-īā-* ‘strength’ > Old Irish *treise īā*, f. ‘strength, vigour’; cf. Jasanoff 1991), as well as for deadjectival verbs. This synchronic rule of suffix deletion is what characterises the Caland system.

There are numerous verbal formations with deadjectival semantics that are associated with the Caland system, mostly characterised present stems involving

¹A comprehensive study of Caland formations in all lexical domains of Celtic will be published in Höfler & Stifter (in preparation).

the suffixes **-eh₁ie/o-*, **(eh₁)skē/o-*, **-ie/o-*, **-éje/o-*, nasal infix presents, and regular thematic presents (cf. Rau 2009: 136–160). Unlike Bozzone (2016), we see no convincing evidence for Caland-associated primary root aorists (see also Section 3.1 of the introduction to this volume), but, with Rau (2013), there seems to have been an association with the perfect. The present stems are attested in all three meanings expected for a deadjectival verb: stative, fientive, and factitive. While some of the involved suffixes are predominantly associated with one of the three meanings (e.g., **-eh₁ie/o-* and stative semantics), this specialization might reflect a later development. In Celtic, some types are attested in more than one meaning.

Celtic preserves only a small, yet exquisite number of Caland verbs. Four of the present stem types listed in the previous paragraph are attested, as well as one Caland-associated perfect. There are two roots for which more than one present stem is found. While there is evidence for a certain amount of productivity (see below), none of these types becomes (or remains) the go-to template for deadjectival verb formation in Celtic.

When collecting the verbal stems, we applied the label “Caland root” relatively generously. As is clear from the introduction to this volume (see especially Section 3.1), it is often not easy to define such a root based on morphological or semantic criteria alone. On the one hand, the typical Caland suffixes in the adjectival and nominal domains also appear with verbal roots (e.g., **ġneh₃-* ‘recognise, know’ and **ġnh₃-ró-* ‘knowing, known’ > Latin *gnārus* ‘knowing, skilled’, Greek γνῶρίζω ‘make known’), and on the other hand, some roots allow both a verbal and an adjectival interpretation, such as **seġ^h-* meaning either ‘overpower, overcome, subdue’ (e.g., in the thematic present **séġ^h-e-t(o)i* > Vedic *sáhate* ‘overcomes’, Greek ἔχει ‘has’, or the agent noun **séġ^h-tor-* > Vedic *sádhar-* m. ‘victor’, Greek ἔκτωρ) or ‘(be/make/become) strong, victorious’ (e.g., in the perfect **se-sóġ^h-e* with an intransitive meaning in Vedic *sāsáha* ‘is victorious’, or the comparative **seġ^h-ios-* > Ved. *sáhyas-* ‘stronger’). In such a case, it is conceivable that adjectival ‘(be/make/become) strong, victorious’ was the original meaning and that a verb **séġ^h-e-toi* ‘is strong, victorious’ (cf. Greek ἐρεύθεται ‘is red’) was construed with an object accusative, from which the meaning ‘is strong, victorious (sc. in relation to somebody or something)’ > ‘overpowers (somebody or something)’ could then have developed. The same pathway can essentially be observed for **d^hers-* ‘(be/make/become) bold, courageous’ in the case of Greek θαρσέω, intransitive ‘be confident’ (*Il.*+) and transitive ‘dare’ (*Od.*+) < ‘be bold/confident in regard to something’ (cf. θάρσει τόνδε γ’ ἄεθλον ‘take on this contest with courage’ *Od.* 8.197) and similarly for Vedic pres. *dhṛṣṇóti*, perf. *dadhárṣa* intransitive ‘be courageous’ and transitive ‘dare’.

This is why we include some roots in our collection that are usually seen as verbal roots (e.g., **h₂eǵd^h-* or **h₁aǵd^h-* ‘burn, ignite’) if typical Caland morphology is attested in Celtic or elsewhere (e.g., **h₁áǵd^h-u-* > Old Irish *aed* ‘fire’, **h₁(a)ǵd^h-ró-* in Greek αἶθρη f. ‘the clear sky’, ἰθαρός ‘serene, clear’, αἰθί-οψ ‘burnt-face’) and if an originally adjectival meaning may therefore be postulated (e.g., ‘(be/make/become) burning, bright’).² We are aware of the potential risk in applying such a broad definition of what a Caland root (and, thus, a Caland verb) is, but we believe that for the sake of completeness and a more comprehensive understanding of the system, it is justified to be more inclusive. In any event, we will address the potential verbal meaning of these ambiguous roots individually in the commentary section of the respective roots and make sure they do not unduly influence our overall conclusions.

In total, Celtic exhibits twenty Caland-associated verbal stems. Eleven of these have cognates in other languages, which means that they are probably inherited. The remaining nine do not have cognates elsewhere and are therefore perhaps inner-Celtic creations. In these cases, it is an interesting question to explore whether there is evidence for a corresponding adjective that could serve as the basis for the verb. Since the verb is nonetheless root-derived, not stem-derived (i.e., lacks the adjectival suffix), this would point to at least some productivity of the Caland system in the verbal domain in Celtic (see Section 7).

2 The ‘stative’ type

2.1 Background

The suffix of the so-called ‘stative’ type (as per Jasanoff 1978, Jasanoff 2004; see also Peters 2007, Kümmel 2012, Malzahn 2014, Peters 2016, Jasanoff 2021) is reconstructed as **-eh₁ǵe/o-* (> **-eǵh₁e/o-* ?) and usually takes a zero-grade root. Structurally, the statives are interpretable as decasuative present stems in **-ǵe/o-* derived from adnominally used instrumental singular forms in **-eh₁* of root nouns, for example **h₁rúd^h-* ‘redness’ (OIr. *rú* f. ‘redness, red dye plant’) → **h₁rud^h-eh₁* ‘with redness; red’ → **h₁rud^h-eh₁-ǵe/o-* (Latin *rubeō* ‘I am, become red’, Old High German *rotēn* ‘flush’).³ If this analysis is correct, the ‘stative’ type would be expected to have typical deadjectival semantics, which means that the stative mean-

²For **h₂eǵd^h-* (**h₁aǵd^h-*) as a Caland root, see Nussbaum (1976: 35): “**h₂eǵd^h-* ‘burn, shine’ [...] makes a well-attested Caland system”.

³LIV² 508–509 sets up an “Essiv” **rud^h-h₁ǵe-* here. For the “Essiv” type see further LIV² 25. Schumacher (2004: 41–42), who otherwise follows LIV² closely, remarks that both Celtic and Latin require **-eh₁-ǵe/o-*, which may be secondarily introduced.

ing ('be red') is only one of three predictable meanings of such a present stem, viz. besides fientive ('become red') and factitive ('make red'; cf. Malzahn 2014: 94, n. 43).

In Celtic, the suffix(oid) **-eh₁-je/o-* develops to Proto-Celtic **-ī-* and is one of the sources of Old Irish W2 verbs with a palatal root final consonant and a 3sg. conjunct ending *-i* (e.g., *ruidid*, *·ruidi* 'flush, blush'). In Middle Welsh, the stative type results in verbs with raising in the root vowel of the 3sg. and an ending absolute *-it* (archaic), conjunct *-Ø*. This type is attested for four Caland roots in Celtic (cf. Schumacher 2004: 41–42), a transitive-factitive meaning is well-represented beside the stative or fientive, though it is not clear whether this is an archaism or an innovation. Three of those four verbs have cognates in other branches of Indo-European.

2.2 **(h₁)reud^h-* 'red'

**(h₁)rud^h-eh₁-je/o-* 'be/make/become red' > PCelt. **rud-ī-* > OIr. *ruidid*, *·ruidi* 'flush, blush; make red' (cf. Schumacher 2004: 552–553).

- (1) Old Irish (*Críth Gablach* 548)

ataat trí toichnedai frisná ruide tocrád
are three fastings at.which.not blush.3SG.PRS.IND vexation
ríg
king.GEN.SG

'There are three fastings at which the offended majesty of a king does not blush.'

- (2) Old Irish (*Anecd.* I 12.26)

a rí ruides gréin
oh king reddens.3SG.PRS.IND.REL sun
'oh king who reddens the sun' (var. lect. *reithes* 'moves')

Cognates Latin *rubēre* 'be/become red', Old High German *rotēn* 'be/turn red', Latvian *rudēt* 'rust', Russian Church Slavonic *rōdēti sja* 'become red' (LIV² 508–509).

Adjective Old Irish *rúad*, archaic and poetic *ród* 'red' < **(h₁)re/oud^h-o-* (cf. Gaul. *roudo-*, OW, OCorn., OBret. *rud*, ModW *rhudd*, MCorn. *ruth*, *ruyth*, NBret. *ruz* 'red'; also Latin *rūfus*, *rōbus*, Proto-Germanic **rauda-*, Lithuanian *raūdas*, Old Church Slavonic *rudŏ*).

Commentary The preponderant, and probably primary, meaning of the verb in Early Irish is fientive ‘turn red’ (namely in the face, i.e., ‘to flush, blush’, usually as a consequence of being shamed), or stative ‘be red’. Possible examples of factitive ‘make red’, as a verb or as a verbal noun, are very rare. They are only found in poetry, unfortunately often in philologically difficult contexts. The form *ruides* in the example cited from the tale *Scéla Cano meic Gartnáin* is attested in the single manuscript copy of the tale. Its varia lectio *reithes* ‘who moves, makes run’ occurs only in the philologically less reliable *Nebenüberlieferung* of the *Mittelirische Verslehren*; the causative meaning ‘move, make run’ of *reithid* is in itself unusual.

2.3 **k^(u)sueib^h*- ‘swift (?)’

**k^(u)suib^h-eh₁-je/o-* ‘be/make/become swift’ > PCelt. **ksuib-ī-* >> MIr. *scibid*, *·scibi* (with sporadic metathesis of the initial cluster?) ‘flow, swim (of a vessel); move; move sth. jerkily’, MW *chwyfu* (3sg. *chwyf*) ‘move to and fro, budge’, MBret. *fifual* ‘move’ (cf. Schrijver 2003; Schumacher 2004: 422–424).

- (3) Middle Welsh (WM 455_{36–8} = CO 80; 14th c.)

Ni **chwy[u]ei** ulaen blewyn arnaw
not move.3SG.IMP tip hair on.him
‘Not a tip of a hair would stir on him.’

Cognates Old High German *swebēn* (amongst other meanings) ‘be agitated (of the sea), be in high and low tide’.

Adjective W *chwith* ‘left, awkward, sinister, wrong’ < **k^(u)suib^h-tó-* (compare English *swift*).

Commentary The basic meaning of the root is not quite clear. It seems to refer to some sort of motion, not a single jerky movement, but rather a state or property or an iterative kind of motion. Based on Caland morphology, especially in Iranian (cf. Young Avestan *xšuuīβra-* ‘quick, vibrant’, *xšuuīβi.išu-* ‘with vibrant arrows’, *xšuaaēβa-* ‘moving quickly’, on which see Schrijver 2003), we assume an adjectival rather than a verbal basic meaning.

A potential Caland adjective **k^(u)suib^h-tó-* (compare English *swift*) is attested in W *chwith* ‘left, awkward, sinister, wrong’, though the meaning is quite removed from the purported meaning of the root. Schrijver (2003) has argued that

ModIr. *ciotach* (the unattested OIr. form would be **cittach*) ‘left-handed, awkward, clumsy’, which could be compared with *chwith* under the assumption of a sporadic, irregular loss of initial *s-*, is unrelated. The Irish verb is found both with forms that point to lenited and unlenited *b*; forms with the latter preponderate in the later period. Besides, in ModIr. the verb *sciobaidh* ‘snatch, sweep away’ is found (eDIL s.v. *scipaid*). The reason for the variation in the status of lenition and palatalisation in these forms is obscure. Contamination with Old Norse *skip* ‘ship’ may have been a factor or sound-symbolism may be at play (unlenited **scipaid* presupposes a geminate **-bb-*, for the sound-symbolic character of which see Stifter 2024, especially Section 11).

2.4 **temk-* ‘hard’

**tṃk-eh₁-je/o-* ‘be/make/become hard’ > PCelt. **tank-ī-* > OIr. *·téici* ‘congeal, become solid; make solid’.

- (4) Old Irish (ZCP 7, 1910. 484.33 = *Enchiridion Sancti Augustini*)
con-<ro>·théced (gl. *adiunctus atque concretus*)
 <AUGM>·**thicken**.3SG.PRET.PASS
 ‘was solidified’

Cognates –

Adjective OIr. *técht* ‘(of liquids) thick, frozen; calm, peaceful’ < **tṃk-to-* (cf. Pashto *tāt* ‘close, thick’).

Commentary McCone (1998: 469–470 and 474–475) and LIV² 625–626 see OIr. *·téici* as the remodeling of a nasal infix present **tṃ-né/n-k-*. Schumacher (2004: 615–616), on the other hand, argues that the continuant of a nasal infix present ought to be OIr. **·tic* and reconstructs a thematic present **tṃk-e/o-* instead. The weak inflection would then be secondary. However, nothing speaks against a regular stative **tṃk-eh₁-je/o-* > PCelt. **tank-ī-* > OIr. *con·téici* (now also Schumacher, p.c.).

The meaning ‘calm, peaceful’ of the adjective OIr. *técht* could be metaphorical and thus secondary within Irish. On the other hand, the connotation ‘established *status quo* < lack of fluidity’ is supported by OW *tagc*, MW *tanc* ‘peace’, Gaul. PN *Tanco-rix* (if ‘peace-ruler’⁴) < **tṃk-o-*.

⁴This provisional translation is agnostic as to what precisely is meant by ‘peace’, a concept

2.5 **tuem*- ‘thick’

**tum-eh₁-iē/o-* ‘be/make/become thick’ > PCelt. **tum-ī-* > MW *tyfu* (3sg. *tyf*) ‘grow; make grow; heal’, MBret. *teñvañ*, *tiñvañ* ‘grow together, develop sap, heal’ (OBret. 3sg. *ni-tum*), MCorn. *tevi* ‘grow’ (cf. Schumacher 2004: 646–648).

(5) Middle Welsh (J 1 1082; c. 1400)

Tyuit *maban*, *ny thyf* *y gadachan*
grow.3SG.PRS.IND infant not grow.3SG.PRS.IND his swaddling-clothes
‘An infant grows, his swaddling-clothes grow not.’

Cognates Latin *tumēre* ‘be/become swollen’, Lithuanian *tumėti* ‘become thick, coagulate’ (LIV² 654 s.v. **tuem-*).

Adjective There is no corresponding simplex adjective, but we find a compound MW *hydwf* ‘full-grown, thriving’, OBret. PN *Hutum* < **su-tum-V-*. It is formally possible to equate these with Ir. *sothaim* ‘amusing, agreeable’, *dothaim* ‘morose, grim, surely’ < **su-/du-tum-i-*, perhaps in the sense ‘has grown/become good/bad’.

Commentary Based on the cognates in Latin and Lithuanian, PCelt. **tum-ī-* can be considered an inherited Caland verb. A *-*ró*-adjective is attested in Vedic (*túmra-* ‘strong, stout’).

3 Nasal infix presents

3.1 Background

Nasal infix presents are a well-attested class in Celtic (cf. McCone 1991; Schumacher 2004: 42–45). Five such present stems are attested from Caland roots, three of which have factitive (and only factitive) meaning. This seems to support the widely held notion that the original function of this present stem class was to create factitives (see, e.g., Rau 2009: 143–146; this volume). Yet, the evidence of the individual languages is ambiguous and might indicate that this need not necessarily have been true in the proto-language. Fientive meaning is attested, for example, in Old Russian *svbnuti* ‘become bright, day’, Lithuanian *švintù* ‘become bright’ (**ķuejt-* ‘bright’; cf. the ‘stative’ Lithuanian *švitėti* ‘shine brightly’),

that is historically and culturally very ambivalent, and whose interpretation in the context of Gaulish society may differ significantly from 21st century ideas of peace.

in Lithuanian *senkù* ‘drop (of water level)’, Old Church Slavonic *i-sečētō* ‘will dry out’ (**sek-* ‘dry’), and stative meaning in Latin *languēō* ‘be slack, languid’, Ancient Greek λαγγών ‘coward’ (**sleg/g-* ‘slack’), or in Ancient Greek ἀνδάνω (+DAT) ‘please, delight’ < ‘be tasty’ (**sueh₂d-* ‘tasty’; cf. the factitive meaning of Vedic *s_(u)vádati* ‘make tasty’ from **suh₂-ŋ-d-* as per LIV² 606, n. 2b).⁵ The remaining two nasal infix presents from Celtic might corroborate this.

3.2 *(*h*₁)*reud^h*- ‘red’

*(*h*₁)*ru-n(e)-d^h*- ‘make red’ >> PCelt. **rund-e/o-* > Old Irish *rondaíd*, *·roind* ‘dye red’, *fo·roind* ‘stain’ (cf. Schumacher 2004: 553–554, LIV² 508–509).

- (6) Old Irish (Ält. Ir. Dicht. i 39 § 3 = Rawl. 115a25)
gáeth rúad rondaíd *for fáebur fuilchiaíd*
 wind red **redden**.3SG.IMPf on edge blood.mist
 ‘A red wind which coloured sword-edges with a mist of blood.’

Cognates –

Adjective OIr. *rúad* ‘red’ < **h₁roud^h-o-* (see Section 2.2 above).

Commentary For the vocalism *o* < **u*, see Stüber (1998: 87). *Rondaíd* and its compound *fo·roind* occur mostly in contexts of fighting and battles with respect to things that get coloured by blood.

3.3 **h*₂*eīd^h*- or **h*₁*aīd^h*- ‘burning, light’

**h*_{2/1}*i-n(e)-d^h*- ‘make burning’ >> PCelt. **ind-e/o-* > MW *enn-ygnu* (3sg. *enn-ygn*) ‘kindle, light a fire’ (cf. Schumacher 2004: 374–375; LIV² Addenda s.v. **h*₂*eīd^h*-).

- (7) Middle Welsh (WML 26; 14th c.)
Ef a enyn *y ganhwyll gyntaf rac* *bron y brenhin*
 he REL **light**.3SG.PRS.IND the candle first before breast the king
 ‘He lights the first candle before the king.’

Cognates Vedic *indhé* ‘lights a fire’ (LIV² 259 s.v. **h*₂*eīd^h*-)

⁵On this phenomenon, see Gorbachov (2007) (for Balto-Slavic and Germanic); Rau (2009: 146, n. 68); this volume.

Adjective –

Commentary This root may be a verbal root (see Section 1 for a discussion). Other Celtic material from this root includes OIr. *áed* *u*, n. ‘fire’, Bret. *oaz* m. ‘rivalry, jealousy’ (sceptical LEIA A-19) and the Gaul. ethnonym *Aeduī* (in Latin transmission). An etymological connection with the root **and-* represented in OIr. *andud* ‘act of kindling, lighting’ and *ad-andai* ‘kindle, light’ (suggested, for example, in LEIA A-75) is formally excluded (Schumacher 2004: 375).

3.4 **pleh₁-* ‘full’

**p_{l̥}-n(e)-h₁-* ‘be/become full’ > ? PCelt. **pli-ni-* > OIr. *do·lin* ‘flow, abound’ (cf. McCone 1991: 18, 21, 34–35; Schumacher 2004: 374–375).

- (8) Old Irish (Sg. 158a1)
do·linim (gl. *māno*)
 flow.1SG.PRS.IND
 ‘I flow.’
- (9) Old Irish (Ml. 56a14; 83c7)
du·linat (gl. *mānantium* and *pollentes*)
 flow.3PL.PRS.IND
 ‘which flow’

Cognates Vedic *prṇāti* ‘fills’, Old Avestan *pərənā* ‘fill’, Latin *polleō* ‘be strong, able’, Armenian *lnwom* ‘I fill’, Albanian *m-blon* ‘fills’ (LIV² 482–483 s.v. **pleh₁-*).

Adjective OIr. *lán* ‘full’ < **plh₁-no-* (cf. W *llawn*, B *leun* ‘full’).

Commentary The root features prominently in Caland-associated nominal formations: **plh₁-no-* > OIr. *lán*, W *llawn* B *leun* ‘full’, **pelh₁-u-* > Gaul. *elu-*, OIr. *il-* ‘many, much’, W *el-*, comp. OIr. *lia* ‘more’, equative *lir* ‘as much as’, or W *llanw* ‘tide, flow, flood, influx’, OBret. *lanu*, NBret. *lanv* ‘flood’ < **planuō-* < **plh₁-n-uō-*. On its status as a verbal or adjectival root, see Nussbaum (2022: 208, n. 7) and Section 3.1 of the introduction to this volume.

Determining the precise inflectional class of *do·lin* ‘flow, abound’ (thus the headword in eDIL dil.ie/18026; later continued as the simple verb *tuilid*) is difficult, as is the demarcation against the verb that is set up in eDIL as *do·lína* ‘increase, add to’ (dil.ie/18028). The 3pl. *do·linat* displays a non-palatalised present-stem-final *n*, which is *a priori* not compatible with the reconstructed stem **pli-ni-*.

It rather points to **φli-nu-*, perhaps with a transfer to the class of *nu*-verbs; unless the alternation in the palatalisation of the stem-final consonant is analogical to the S1 verbal class, the old *e/o*-thematic verbs. The augmented 3sg. preterite *do·rulin* gl. *manasse* (Ml. 64c18) shows that the nasal was synchronically felt to be part of the root, as if it were a weak verb. To complicate matters, the root of OIr. *do·lin* may alternatively be from or be influenced by **leǵh_x-* ‘pour’ (MW 3sg *dillyǵ* ‘flows’; see below 4.5), or a merger of both roots may have occurred (Stifter 2023: 176–177). If this is the case, and if any credence can be given to the synchronic behaviour of the verb in Old Irish, the laryngeal of that root may be specified as **h₃*, i.e., PCelt. **linú-* < **linō-* < **li-n-e-h₃-*.

3.5 **seh₁(i)-* ‘long’

**seh₁-n(e)-i-* ‘make long’ > PCelt. **sī-ni-* >> **sīnī-* > OIr. *sínid*, *·sín(i)* ‘stretch (one-self), extend’.

(10) Old Irish (LU 4927)

sínithi *iarom co* *memdatar* *in da liic*
stretch.3SG.PRS.IND.himself then so.that break.3PL.PRET the two stones
ro *batár immi*
 AUGM were around.him
 ‘He stretches himself so that the two stones around him broke.’

Cognates –

Adjective OIr. *sír* ‘long’ < **seh₁-ro-* (cf. W, Corn., Bret. *hir* ‘long, tall, lengthy’).

Commentary The verb is only attested in sources from the Middle Irish period onwards. All attestations are compatible with the analysis as a weak W2a verb. This implies that the inherited strong nasal-infix verb was re-interpreted as a weak *ī*-verb at some point in the prehistory of Irish. The root features prominently in Caland-associated nominal formations: comparative OIr. *sia*, *sía*, MW *hwy* ‘longer’, superlative OIr. *síam*, MW *hwyaf*, OBret. *huiam* ‘longer’, all of which demonstrate the absence of the adjectival suffix **-ro-* vis-à-vis the underlying adjective **seh₁-ro-* > OIr. *sír* ‘long’, W, Corn., Bret. *hir* ‘long, tall, lengthy’.

3.6 **uelh_x*- ‘powerful, in power’

**ul-n(e)-h_x*- ‘be powerful’ >> PCelt. **ual-na-* >> **uall-nā-* > OIr. *follnathir* ‘rule’ (cf. Schumacher 2004: 655–656).

(11) Old Irish (Tec. Corm. §2)

recht fallnathar *for talman* *tuind*
right rule.3SG.PRS.IND.REL over earth.GEN.SG surface.DAT.SG
‘The right that rules on the surface of the earth.’

Cognates –

Adjective **ulh_x-o-* ‘powerful, in power’ as second compound member **-ualo-* in Gaul. *Katouualos*, OW *Catgual*, OIr. *Cathal* (**katu-ualo-* ‘battle king’ *vel sim.*) or PrimIr. *Cunouali*, Ogam *CUNAVA[LI]*, archaic OIr. *Conual*, OIr. *Conall* < **kuno-ualo-* ‘hound king’.

Commentary Other Caland material belonging to this root: Latin *ualeō* ‘am strong, have power’, *ualidus* ‘strong, healthy’, Toch. B *walo*, A *wāl* ‘king’ (< **ulh_x-ont-*). The majority of verbal stems attested in the Indo-European languages point to a verbal meaning ‘to rule’, however (see LIV² 676–677 s.v. 1. **uelH-* ‘stark sein, Gewalt haben’). According to McCone (1991: 16), the OIr. inflection speaks in favour of setting up the root-final laryngeal as **h₂* (see also LIV² 676 s.v. 1. **uelH-*, n. 1).

It is doubtful if the rare glossatorial and poetic word OIr. *²fāl* ‘king’ (eDIL dil.ie/21232) continues the uncompounded substantivised adjective **ulh_x-o-* ‘powerful, in power’. Its long vowel, which is secured by rhyme, is incompatible with the proposed etymology. Perhaps *²fāl* ‘king’ is only a figurative use of *¹fāl* ‘wall’.

A number of sound developments in the OIr. verb are unexpected. The preponderant *o* of the initial syllable of *follnathir* in Old Irish, beside occasional *fallnathir*, is probably due to sporadic rounding of **a* caused by the preceding labial consonants. The vocalisation PCelt. **ualn-* instead of **ulin-* < **uln-* is probably based on the influence of **ualo-* < **ulh_x-o-*. The *-lln-* of *follnathir* could either be the result of contamination with *follán* ‘sound, full’, or due to the regular change of **-ln-* > **-ll-* (Stifter 2024: Section 2.1; cf. 3pl. pass. *follatar* Ml. 77b4), with subsequent reintroduction of the present-tense morpheme *-n-*.

If the interpretation as a Caland nasal infix present is correct, the stative meaning ‘be powerful’ (instead of factitive ‘make powerful’) is noteworthy.

4 Plain thematic presents

4.1 Background

Plain thematic presents typically have a full grade in the root and are attested in stative and fientive (active or middle) and, less frequently, in factitive meaning (active) (see Rau 2009: 146–157; Rau 2013: 258–265; Villanueva Svensson 2021). Compare Vedic *rócate* ‘shines’, Young Avestan act. ptcpl. *raociṇt-* ‘shining’, Tocharian B *lyuštär* ‘illuminates’ (: **leuk-* ‘bright’), Vedic *tápati/tápate* ‘be hot, burn’, *tápati* ‘make hot, burn (sth.)’, Khotanese mid. *ttavāre* ‘are hot’ (: **tep-* ‘hot’), or Greek ἄγγω ‘squeeze, throttle’, Latin *angō, -ere* ‘bind, throttle’ (: **h₂emĝh-* ‘narrow’).

In Celtic (cf. Schumacher 2004: 36–37), there are two verbs with a full-grade root and two verbs with zero-grade root (one of which is uncertain). Rather than being continuants of *tudāti* presents, the latter owe their root vocalism to analogical influence from nominal formations that (in both cases) have a zero grade throughout. Two of these verbs are factitive, one is stative/fientive, and one is stative; only one has cognates in other Indo-European branches.

4.2 **g^{uh}er-* ‘hot’

g^{uh}ér-e/o-* ‘make hot’ > PCelt. **g^uer-e/o-* > OIr. *geirid, *fo-geir* ‘heat, inflame; irritate’ (cf. Schumacher 2004: 372–373).

(12) Old Irish (Metr. Dinds. iv 334)

aní fo-geir mo menma dom-beir fri degra
 the.DEICTIC heat.3SG.PRS.IND my spirit brings.me to *degra*
duibe.
 darkness.GEN.SG

‘The thing that frets my spirit brings me to the *degra* (fire?) of darkness.’

Cognates Ancient Greek *θέρπομαι* ‘become warm/hot’, *θέρω* ‘heat’,⁶ Alb. *zien* ‘cook (tr.)’ (LIV² 219–220).

Adjective OIr. *gor* ‘pious’ < **g^{uh}or-o-* (cf. Bret. *gor* ‘warm’, W *gwâr* ‘civilised, gentle, kind’).

⁶In this type of Greek verbs, the middle inflection with intransitive meaning is often attested earlier than the active inflection with transitive meaning (cf. *θέρπομαι* ‘become warm/hot’ Hom.+ : *θέρω* ‘heat’ A.R.+); see Rau (2009: 155).

Commentary The simple verb *geirid* is only attested in a single Middle Irish text, in the form *gerthiut* ‘it warms thee’ (BDD² 317). This is most probably an analogically created form.

4.3 **th₂eys-* ‘silent’

**th₂eys-e/o-* > ‘be/become silent’ > PCelt. **taus-e/o-* > OIr. *tóaid* ‘be silent’, remodeled as Late Proto-British **tau-i-* > MW *tewi* (1sg. *tawaf*, 3sg. *teu*) ‘be/become quiet, silent’, MBret. *teuell* ‘be/become quiet, silent’, MCorn. *tewel* ‘be quiet, silent’ (cf. Schumacher 2004: 621–623 without *tóaid*).

- (13) Old Irish (*Ériu* iv 30.22)

no thoad in sluag

PTCL **be.silent.3SG.IMPF** the host

‘The host was/became silent.’

- (14) Middle Welsh (GMB 76; 12th c.)

Py dawant anant, na phrydant

why **be.silent.3PL.PRS.IND** musicians that.not compose.3PL.PRS.IND

wawd?

praise

‘Why are the musicians silent that they do not sing praise?’

Cognates –

Adjective OIr. *tó* < **th₂e/ous-o-* and *tóe*, *túae* < **th₂e/ous-īo-*, both ‘silent’.

Commentary OIr. *tóaid* may continue the primary verb or be a secondary, denominational formation from OIr. *tó* ‘silent’. The interjection OIr. *tá* ‘quiet! hush!’ has been tentatively explained as the outcome of the imperative **th₂eys-e!* (cf. LEIA T1–2), but this is phonologically implausible.

4.4 **meh₂k-* ‘big’

**mh₂k-e/o-* ‘make big’ > PCelt. **mak-e/o-* > OIr. (*do-for*)*maig* ‘increase, add’, MW *magu* (3sg. *mac*) ‘rear, breed, nourish, foster’, MBret. *maguaff* ‘feed, nourish, rear, suckle’, MCorn. *maga* ‘rear, nourish’ (cf. Schumacher 2004: 466–470).

- (15) Old Irish (Fél. Ep. 195)
do-formaig *a ánae*
 increase.3SG.PRS.IND his wealth
 ‘It increases his wealth.’
- (16) Middle Welsh (MA² 241_{a47-8}; 12th–13th c.)
or a fag magu *daear hi*
 if REL feed.3SG.PRS.IND earth her
 ‘if the earth nourishes her’

Cognates –

Adjective Old Irish *machtae* ‘great’, MW *meith* ‘long, large’ < **mh₂kt(i)io-*; MW *maeth* m. ‘sustenance, nourishment, fosterage’ < **mh₂k-to-* (a substantivised adjective?).

Commentary OIr. *machtae* ‘great’, MW *meith* ‘long, large’ may originally have been the participle (**mh₂kt(i)io-*) of the verb PCelt. **mak-e/o-* ‘raise, increase’, but both phonologically as well as semantically these forms and MW *maeth* could also belong with the root **meġ-* ‘big, great’ qua **m_aġt(i)io-* and **m_aġ-to-*, respectively.

4.5 **lih_x-* ‘liquid (?)’

**lih_x-e/o-* ‘be liquid (?)’ > PCelt. **liġ-e/o-* > MW 3sg. *dillyð* ‘flow, abound’ (cf. Schumacher 2004: 372–373).

- (17) Middle Welsh (T 21₂₄; 14th c.)
Gogwn pan dillyd, *gogwn pan wescryd.*
 I.know when flow.3SG.PRS.IND I.know when ebb.3SG.PRS.IND
 ‘I know when it flows, I know when it ebbs away.’

Cognates –

Adjective Not attested directly, but the following nouns may be substantivisations of Caland adjectives: W *llin* ‘flow of blood, pus, discharge’ < **lih_x-nó-*, OIr. *liē* ‘flood, spate’, W *lli*, pl. *lliant* ‘flood(s)’ < **lih_x-nt-*, OIr. *ler*, W *llŷr* ‘sea, ocean’ < **lih_x-ró-*.

Commentary Despite the evidence for a set of typical Caland-derivatives (see above), the root **lejh_x-* may be a non-Caland verbal root with a meaning ‘to pour’ (cf. LIV² 405–406). See also Section 3.4 on the question of the merger of this root with **pleh₁-* ‘fill’.

5 Causatives/iteratives

5.1 Background

The causatives/iteratives are characterised by an *o*-grade in the root and the suffix **-éje/o-*, and also feature among the typical Caland present stems (cf. Rau 2009: 141–143). It is disputed whether they constitute a veritable root-derived verbal stem type in the proto-language or if they are rather denominal verbs derived from **R(ó)-o-* verbal nouns or adjectival abstracts (e.g., **g^{uh}óro-* ‘heat’ → **g^{uh}or-e-je/o-* ‘apply heat’⁷).⁸ These two viewpoints are not necessarily mutually exclusive. A set of frequent denominal verbs could be seen as the diachronic source of a subsequently abstracted verb-formation type, which then encouraged the creation of new root-derived verbal stems by formal analogy.

As root-derived, true causatives, we would expect factitive semantics to be predominant. If they were originally denominal verbs based on an adjectival abstract ‘X-ness’, however, all three possibilities (stative, fientive, factitive) may be expected, although a secondary specialization of one of them would not be surprising. Within the Indo-European languages, the factitive type prevails, with some notable exceptions. Compare the polysemy of Latin *lūcet* (if **loukejeti*) ‘is light, is clear, shines’ and ‘lights (sth.)’, which, however, might be due to a conflation with the zero-grade stative.

In Celtic, this type is attested for six roots, four of which have extra-Celtic cognates. Originally, the meaning of all of them was likely factitive, though some of the attested verbs have undergone substantial semantic change. An *o*-grade thematic stem is attested for three of them (one is uncertain), and could be the derivational basis in a not-so-distant past for at least two of them.

5.2 **b^herg^h-* ‘high’

**b^horġ^h-eje/o-* ‘make high’ > PCelt. **borg-ī-* > OIr. *·dibairg* (suppletive prototonic stem of *do·bidci*) ‘throw, shoot, hurl’, MW *bwrw* ‘throw, cast, strike’ (cf. Schrijver 1995: 55–56; Schulze-Thulin 2001: 120–121).

⁷It is also thinkable that the basis was the instr. sg. of the underlying noun: **g^{uh}óreh₁* ‘with heat’ → **g^{uh}or-eh₁-je/o-* (> **g^{uh}or-ejh₁e/o-?*) ‘be/become/make with heat’.

⁸On this question, see Peters (2016); noncommittal Bozzone (2020).

- (18) Old Irish (RC iii 178 (LL))

ro-díbaire *inn gai dó*
 AUGM.**throw**.3SG.PRET the spear at.him
 ‘He hurled the spear at him.’

- (19) Middle Welsh (LIDW 63₁₀; 13th c.)

o buru *y mor betheu yr tyr neu yr traeth hunnv*
 if **throw**.3SG.PRS.IND the sea things to.the land or to.the beach that
byeuyt y brenhyn
 owns the king
 ‘But if the sea throws anything on the land or on that beach, they belong
 to the king.’

Cognates Vedic (*sám*) *barháya-* ‘strengthen’, YAv. (*us*) *barəzaia-* ‘rear’ (LIV² 78–79).

Adjective **b^hrg^h-(é)nt-* ‘high’ (see Section 1), but semantically distant from the synchronic meaning of the verbs.

Corresponding o-grade thematic stem –

Commentary The semantic development underlying the Celtic verbs can be imagined as ‘make high’ > ‘raise’ > ‘throw’ and must predate the diversification of the Insular Celtic languages.

5.3 **g^{uh}er-* ‘warm, hot’

**g^{uh}or-eje/o-* ‘make warm, heat’ > PCelt. **g^uor-ī-* > OIr. *guirid* ‘warm, hatch’, MW *gori* ‘brood, hatch, fester’ (cf. Schulze-Thulin 2001: 146–147; LIV² 219–220).

- (20) Old Irish (Ml. 39c24)

guirit *són (gl. fouent)*
warm.3PL.PRS.IND this
 ‘they warm’

- (21) Middle Welsh (AL i. 734; 13th c.)

teithi iar yw dodwi a gori
 traits hen is lay.eggs.vN and **brood**.vN
 ‘The nature of a hen is it to lay eggs and to brood.’

Cognates Alb. *n-xeh* ‘warms’ (?) (LIV² 219–220).

Adjective See Section 4.2 above.

Corresponding o-grade thematic stem OIr. *gor o m.* ‘inflammation, pus; the brooding of hens’, W *gôr* ‘pus; hatching’, MBret. *gor* ‘pus, furuncle’ (**g^{uh}or-o-*).

Commentary The evidence is somewhat ambiguous: Since the continuants of the thematic noun do not mean ‘warmth, heat’ any longer, the meaning ‘to warm’ of OIr. *guirid* points to an inherited formation (cf. Alb. *n-xeh* ‘warms’). On the other hand, the ‘brood, hatch’ semantics that are attested for the verbs and the thematic verbal nouns in both Irish and Welsh may suggest an origin as a denominal verb.

5.4 **h₂eĵ* ‘sharp, pointed’

**h₂ok̑-eĵe/o-* ‘make sharp, pointed’ > PCelt. **ok-ĩ-* > MW *hogi* (3sg. *hyc*) ‘sharpen, whet’, NBret. *koñvog*, *kougañ* ‘sharpen, roughen’ (cf. Schumacher 2000: 158 and Schulze-Thulin 2001: 171–173).

(22) Middle Welsh (B XXII 121; 15th c.)

ef a ddaw baedd ac a hyc y ddannedd
PTCL REL come.3SG.FUT boar and REL sharpen.3SG.PRS.IND its teeth
‘There comes a wild boar and sharpens its teeth.’

Cognates Old High German *eggen* ‘harrow’ (cf. LIV² 261).

Adjective OIr. *ér* ‘noble, great’ < **h₂eĵ-ro-* (cf. Ancient Greek ἄκρος ‘utter, highest’, Old Lithuanian *ašras* ‘sharp’), but semantically distant.

Corresponding o-grade thematic stem W *og* ‘harrow’ < **h₂oko-* or **h₂okeh₂-* (but semantically distant).

Commentary The meaning of the verbs in Welsh and Breton is clearly factitive. NBret. *koñvog*, *kougañ* ‘sharpen, roughen’ corresponds to W *cyfogaf*, *cyfogi* ‘make keen or pointed, roughen’. The word-initial *h-* in Welsh is unetymological (Schumacher 2000: 158).

5.5 **temk-* ‘hard’

**tomk-eje/o-* ‘make hard’ > PCelt. **tonk-ī-* > OIr. *tocaid* ‘destine’ (via Proto-Goidelic **toggī-*), MW *tynghu* ‘destine, affirm’, MBret. *tonquaff* ‘destine’ (cf. Schumacher 1995; Schulze-Thulin 2001: 158).

- (23) Old Irish (Thes. ii 293.6–7)

ma rom·thoicther-sa *in so ...*
 if AUGM.me·**destine**.3SG.PRS.SUBJ.PASS-me the this
manim·rotchaither
 if.not.me·AUGM.**destine**.3SG.PRS.SUBJ.PASS
 ‘if this be destined for me ... if it not be destined for me’

- (24) Middle Welsh (R 1036.43, CL1H II 21.a; c. 1400)

truan a dynghet a dynghwyt y lywarch
 miserable of destiny REL **destine**.IMPERS.PRET to Llywarch
 ‘It was a wretched destiny that was destined to Llywarch.’ (Schumacher 1995: 53)

Cognates Old Saxon *ā-thengian*, *an-thengian* ‘carry out, complete’ (cf. LIV² 625–626).

Adjective OIr. *técht* ‘(of liquids) thick, frozen; calm, peaceful’ < **tmk-to-* (but semantically distant). See Section 2.4 above.

Corresponding o-grade thematic stem –

Commentary The expected OIr. outcome of Proto-Goidelic **toggī-* is **tucaid* (thus also Schumacher 1995). The actually attested forms *tocaid* etc. owe their vocalism probably to influence from the verbal abstract *tocad* ‘fortune’ < **tomk-eto-*. The Brythonic verbs have often been treated as belonging to PCelt. **tung-e/o-* ‘swear’ (OIr. *tongaid*, W *tyngaf*, *tyngu*, MBret. *toeaff*). However, Schumacher (1995) has shown that they have to be kept separate.

5.6 **uelk-* ‘wet’

**uolk-eje/o-* ‘make wet’ > PCelt. **uolk-ī-* > OIr. *folcaid* ‘wash the head’, MW *golchi* (3sg. *gwylch*) ‘wash’, MCorn. *gol(g)hy*, MBret. *guelchiff*, NBret. *gwalc’hañ* (see Schulze-Thulin 2001: 141; 159–160; LIV² 679).

- (25) Old Irish (Sg. 146b3)

folcaimm (gl. *lavo*)

wash.1SG.PRS.IND

‘I wash’

- (26) Middle Welsh (T 41, 6–7; 14th c.)

golchettawr *ei lestri, bid groyw ei vrecci*

wash.IMPERS.IMPV his vessels be.3SG.IMPV clear his wort

‘Let his vessels be washed, his wort be clear.’

Cognates –

Adjective PIE **u̯lk-u-* > PCelt. **ulikū-*. This may be continued directly in OIr. *fliuch* ‘wet’. OW *gulip*, ModW *gwlyb*, fem. *gwleb* ‘wet’, Corn. *gleb*, OBret. *gulip*, NBret. *gleb* looks like a thematisation **ulikūo-* > **ulik^uo-* from this (cf. Kümmel 2022).

Corresponding o-grade thematic stem The corresponding noun is **u̯ólk-o-* > OIr. *folc* o, m. ‘heavy rain, wet weather’, perhaps W *golch* ‘a washing, cleansing’ (cf. Latvian *vaļks* m. ‘creek, brook, small stream of water’). Since W *golch* is attested rather late, it is probably a backformation and not a descendant of **u̯ólk-o-*.

Commentary The root **uelk-* (cf. ?**uelk-* in LIV² 679 ‘feucht sein/werden’) is a doublet of **uelg-* ‘wet’ (cf. 2.**uelg-* in LIV² 676 ‘feucht werden’). For the latter, a causative **uolg-eje/o-* ‘make wet’ is attested in Serbo-Croatian *vlāžiti* ‘moisten’, Russian *volóžit* ‘id.’, Lithuanian *válgyti* ‘eat’. It is formally conceivable that **uelk-* is a regular Celtic development of **uelg-sk̑-*. In this case the Caland behaviour of this neo-root **uelk-* would have to be an inner-Celtic development. However, a root-final voiceless **k* is also necessary to explain Lith. *valkà* f. ‘puddle’, Latv. *vaļka* ‘creek, brook, wet meadow’.

5.7 **meg-* ‘big’

mog-eje/o-* ‘make big’ > PCelt. **mog-i-* > OIr. **mogaid/moigid, ·mogai/·moigi (?) ‘make great, increase, magnify’ (> *móaigid*), MW *moi* ‘to foal’ (cf. Joseph 1987: 136; Schumacher 2004: 469).

- (27) Old Irish (*Ériu* xvi 65.47, *Celtica* v 136)
rod-bet o gallaib bann[ach]aib(?) barca ...
AUGM.you-be.3PL.PRS.SUBJ from foreigners zealous ships
maínib mogatar (ms. mogetair)
treasures magnify.3PL.PRS.IND.PASS.REL
‘From industrious (?) foreigners may you have (receive) ... boats that are
magnified (enriched) with treasures.’
- (28) Middle Welsh (*WM* 31₁₋₄; 14th c.)
a phob nos Calanmei y moei, ac ny wybydei neb
and every night 1st.of.May PTCL foal.3SG.IMPF and not knew anyone
un geir e wrth y hebawl.
one word about her colt
‘And every May-eve [the mare] foaled, but no one knew a word about her
colt.’

Cognates –

Adjective OIr. *maige* ‘great’, Gaul. *Magio-rix*, *Magius* < **m*₂*ǧ-iō-*.

Corresponding *o*-grade thematic stem —

Commentary Most attestations of this verb in OIr. are philologically difficult. Variants in *mog-*, *moig-* and *moaig-* or *móaig-* occur seemingly in free variation in the manuscript tradition of texts. *Moaig-/móaig-* could be a reanalysis or even a recent secondary formation based on the comparative *mó* (of *már*, *mór*) ‘bigger, greater’ + the productive verbal suffix *-aig-*. Furthermore, a grave shadow of doubt is cast on the inherited character of this verb by the fact that the regular outcome of PCelt. **mog-ī-* ought to be ***mugaid* in Old Irish (cf. PCelt. **logī-* ‘lay’ > *do-lugai* ‘forgive < *‘lay down’). This is perhaps an indication that the vowel of the first syllable is long in the first place.

The meaning of MW *moi* 'to foal', on the other hand, is difficult to reconcile with 'to make big'. In short, the cognacy between the two verbs is just as uncertain as their derivation from **moġ-eie/o-* 'make big'.

6 Perfects: **temh*_x- ‘faint, dark’

**te-tomh_x*- ‘has become faint’ > PCelt. **tetom*- > **tetam*- > OIr. *tathaim*, *tathamair* ‘died’ (cf. Schumacher 2004: 68–79; 634–635).

(29) Middle Irish (Metr. Dinds. iv 34.52)

conid 'sin *tsléib* **tathamair**
so.that.it.is on.that mountain **die**.3SG.PRET

'So that it is on that mountain that he died.'

Cognates Vedic *tatāma* 'fainted' (cf. LIV² 624; LIV² Addenda s.v. **temH-*).

Adjective OIr. *tem*, *teim* 'dark' < **temh_x-o-*.

Commentary On perfects and the Caland system, see Rau (2013). Besides 'dark' (cf. OIr. *tem*, *teim* 'dark'), the root also had a meaning '(physically) faint' (cf. IEW 1063–1064 "geistig benommen, betäubt" and "dunkel"; LIV² 624 "ermatten, ohnmächtig werden"); compare English *to black out* 'to faint'.

According to Schumacher (2004: 634–635), the irregular vocalism in the reduplication syllable can be explained as based on an unattested present **taimid*, **-taim* 'dies' (cf. pres. *cainid* 'sings' and pret. *cachain*); alternatively, it can have been generalised from the weak stem **tetmh_x-*, which regularly yielded PCelt. **tetam-*. The earliest, albeit Middle Irish, attestations of the verb have deponent endings, perhaps influenced by other suffixless preterites of roots in *-m* such as *lámair* (*ro-laimethar* 'dare') and *dámair* (*daimid* 'endure, suffer').

7 Discussion

To find out whether the derivation of root-based deadjectival verbal stems remained at least to some extent productive in the prehistory of the Celtic languages, it will first be necessary to divide the total number of twenty verbal stems into those with extra-Celtic cognates (A) and those without extra-Celtic cognates (B). Group (A) consists of eleven verbs.

- (A) 1. *(*h*₁)*rud^h-eh₁-je/o-* 'be/make/become red' > PCelt. **rud-ī-* > OIr. *ruidid*, *-ruidi* 'flush, blush; make red' (cf. Latin *rubēre* 'be/become red', Old High German *rotēn* 'be/turn red', Latvian *rudēt* 'rust', Russian Church Slavonic *røděti sja* 'become red')
2. **k^(u)suib^h-eh₁-je/o-* 'be/make/become swift' > PCelt. **ksuib-ī-* > MIr. *scibid*, *-scibi* 'flow, swim (of a vessel); move; move sth. jerkily', MW *chwyfu* (3sg. *chwyf*) 'move to and fro, budge', MBret. *fifual* 'move' (cf. Old High German *swebēn* 'be agitated (of the sea), be in high and low tide')

3. **tum-eh₁-ie/o-* ‘be/make/become thick’ > PCelt. **tum-ī-* > MW *tyfu* (3sg. *tyf*) ‘grow; make grow; heal’, NBret. *teñviñ* ‘grow, increase’ (OBret. 3sg. *ni-tum*), MCorn. *tevi* ‘grow’ (cf. Latin *tumēre* ‘be(come) swollen’, Lithuanian *tumėti* ‘become thick, coagulate’)
4. **h₂/i-n(e)-d^h-* ‘make burn’ >> PCelt. **ind-e/o-* > MW *enn-ygnu* (3sg. *enn-ygnn*) ‘kindle, light a fire’ (cf. Vedic *indhé* ‘lights a fire’)
5. **p_l-n(e)-h₁-* ‘make full’ > ? PCelt. **phi-ni-* > OIr. *do-lin* ‘flow, abound’ (or stative/fientive ‘be/become full’?) (cf. Vedic *prñāti* ‘fills’, Old Avestan *pārənā* ‘fill!’, Latin *polleō* ‘be strong, able’, Armenian *lnwom* ‘I fill’, Albanian *m-blon* ‘fills’)
6. **g^{uh}ér-e/o-* ‘make hot’ > PCelt. **g^uer-e/o-* > OIr. *geirid**, *fo-geir* ‘heat, inflame; irritate’ (cf. Ancient Greek *θέρομαι* ‘become warm/hot’, *θέρω* ‘heat’, Alb. *zien* ‘cook (tr.)’)
7. **b^horĝh-eje/o-* ‘make high’ > PCelt. **borg-ī-* > OIr. *díbaire* (suppletive prototonic stem of *do-bidci*) ‘throw, shoot, hurl’, MW *bwrw* ‘throw, cast, strike’ (cf. Vedic (*sám*) *barháya-* ‘strengthen’, YAv. (*us*) *barəzaia-* ‘rear’)
8. **g^{uh}or-eje/o-* ‘make warm, heat’ > PCelt. **g^uhor-ī-* > OIr. *guirid* ‘warm, hatch’, MW *gori* ‘brood, hatch, fester’ (cf. Alb. *n-xeh* ‘warms’ ?)
9. **h₂ok̑-eje/o-* ‘make sharp, pointed’ > PCelt. **ok-ī-* > MW *hogi* (3g. *hyc*) ‘sharpen, whet’, NBret. *koñvog*, *kougañ* ‘sharpen, roughen’ (cf. Old High German *eggen* ‘harrow’)
10. **tomk-eje/o-* ‘make hard’ > PCelt. **tonk-ī-* > OIr. *tocaid* ‘destine’ (via Proto-Goidelic **toggī-*), MW *tyngthu* ‘destine, affirm’, MBret. *tonquaff* ‘destine’ (cf. Old Saxon *ā-thengian*, *an-thengian* ‘carry out, complete’)
11. **te-tomh_x-* ‘has become faint’ > PCelt. **tetom-* >> **tetam-* > OIr. *tathaim*, *tathamair* ‘died’ (cf. Vedic *tatāma* ‘fainted’)

The existence of corresponding verbal stems in the other branches suggests an inherited status of the eleven verbs of group (A). Note that items 4. and 5. might belong to verbal roots ‘to ignite’ and ‘to fill’, respectively.

Group (B) contains nine verbs, none of which has exact correspondences in the other Indo-European branches. This may suggest that these verbs were created within Celtic. Interestingly, all of them have adjectives (or potential substantivations of such adjectives) beside them.

- (B) 1. **tṃk-eh₁-iē/o-* ‘be/make/become hard’ > PCelt. **tank-ī-* > OIr. *con·téici* ‘congeal, become solid; make solid’ (cf. OIr. *técht* ‘(of liquids) thick, frozen; calm, peaceful’ < **tṃk-to-*)
2. **(h₁)ru-n(e)-d^h-* ‘make red’ >> PCelt. **rund-e/o-* > OIr. *rondaid, ·roind* ‘dye red’, *fo·roind* ‘stain’ (cf. OIr. *rúad*, archaic and poetic *ród* ‘red’ < **(h₁)re/oūd^h-o-*)
3. **seh₁-n(e)-i-* ‘make long’ > PCelt. **sī-ni-* >> **sīnī-* > OIr. *sínid, ·sín(i)* ‘extend, stretch (oneself)’ (cf. OIr. *sír* ‘long’ < **seh₁-ro-*)
4. **u_l-n(e)-h_x-* ‘be powerful’ > PCelt. **u_{al}-na-* >> **u_{all}-nā-* > OIr. *follnaithir* ‘rule’ (cf. **u_lh_x-o-* ‘powerful, in power’ as the second compound member **-u_{alo}-* in Gaul. *Katouualos*, OW *Catgual*, OIr. *Cathal*, etc.)
5. **th₂eys-e/o-* > ‘be(come) silent’ > PCelt. **taus-e/o-* > OIr. *tóaid* ‘be silent’, >> Late Proto-Brit. **tau-i-* > MW *tewi* (1sg. *tawaf*, 3sg. *teu*) ‘be(come) quiet, silent’, MBret. *teuell* ‘be(come) quiet, silent’, MCorn. *tewel* ‘be quiet, silent’ (cf. OIr. *tó* ‘silent’ < **th₂e/oys-o-*; OIr. *túae* ‘silent’ < **th₂e/oys-iō-*)
6. **mh₂k-e/o-* ‘make, become big’ > PCelt. **mak-e/o-* > OIr. *(do·for)maig* ‘increase, add’, MW *magu* (3sg. *mac*) ‘rear, breed, nourish, foster’, MBret. *maguaff* ‘feed, nourish, rear, suckle’, MCorn. *maga* ‘rear, nourish’ (cf. OIr. *machtae* ‘great’, MW *meith* ‘long, large’ < **mh₂kt(i)io-*; MW *maeth* m. ‘sustenance, nourishment, fosterage’ < **mh₂k-to-*, a substantivised adjective?)
7. **lih_x-e/o-* ‘be liquid (?)’ > PCelt. **li_j-e/o-* > MW 3sg. *dillyð* ‘flow, abound’ (cf. the potential substantivisations of Caland adjectives: W *llin* ‘flow of blood, pus, discharge’ < **lih_x-nó-*, OIr. *lië* ‘flood, spate’, W *lli*, pl. *lliant* ‘flood(s)’ < **lih_x-nt-*, OIr. *ler*, W *llÿr* ‘sea, ocean’ < **lih_x-ró-*)
8. **uolk-e_je/o-* ‘make wet’ > PCelt. **uolk-ī-* > Old Irish *folcaid, ·folcai* ‘wash the head’, MW *golchi* (3sg. *gwylch*) ‘wash’, MCorn. *gol(g)hy*, MBret. *guelchiff* (cf. OIr. *fliuch* ‘wet’ < PCelt. **u_{li}ku-* < **u_lk-u-*; OW *gulip*, ModW *gwlyb*, fem. *gwleb* ‘wet’, Corn. *gleb*, OBret. *gulip*, NBret. *gleb* < **u_{li}ku^u-* < **u_{li}kuo-*)
9. **mog^ê-e_je/o-* ‘make big’ > PCelt. **mog-ī-* > OIr. **mogaid/moigid, ·mogail·moigi** ‘make great, increase, magnify’ (>> *móaignid*), MW *moi* ‘to foal’ (cf. OIr. *maige* ‘great’, Gaul. *Magio-rix*, *Magius* < **m₂g^ê-iō-*)

Caution is advised regarding items 4., 7., 9. as they are beset with philological, phonological, and morphological difficulties. In addition, 4. and 7. may have a verbal root at their core. Item 8., the causative/iterative **uolk-eje/o-* ‘make wet’ > ‘wash’, not only has an adjective (**uolk-u-* ‘wet’) but also an *o*-grade thematic stem **uólk-o-* (> OIr. *folc o*, m. ‘heavy rain, wet weather’) beside it, which may have served as the derivational base for a denominal (rather than a deadjectival) verb at some point in the common prehistory of the Insular Celtic languages.

While it cannot be excluded that each of the verbs in group (B) stems from a distant past and once had extra-Celtic cognates that happen to have been lost in the non-Celtic branches of Indo-European, it is also possible to assume that these verbs were created within Celtic based on the corresponding adjective. The five secure examples, items 1., 2., 3., 5., 6., consist of one stative, two nasal-infix presents, and two plain thematic presents. The causative/iterative 8. is ambiguous (see the paragraph above).

Under such an interpretation, it is necessary to conclude that at this point in time, the language could still derive deadjectival verbs from roots and not from adjectival stems, as the respective adjectival suffixes (e.g., **-to-*, **-o-*, **-ro-*) are seemingly deleted and the generated verbs follow the primary formation patterns of deradical verbs. In other words, these examples give the impression that Caland morphology continued to be productive to a limited extent in Celtic. Note that this analysis is possible even for the problematic items 7. and 9.; for the third one, 4. OIr. *follnaithir* ‘rule’, it is necessary anyway to assume that the root vocalism was secondarily adjusted to its potential derivational base **uľh_x-o-* ‘powerful, in power’ > **u_oalo-* (attested as a second compound member), as discussed in Section 3.6.

Support for the assumption of such a limited vitality of Caland morphology in Celtic in general, and a suffix deletion rule in deadjectival derivation in particular comes from the nominal domain. Here, there is evidence for at least two inner-Celtic neo-Caland roots. The first example is the adjective OIr. *trén* ‘strong’ < **treksno-* (cf. also the personal names Gaul. *Trenus*, Ogam *TRENA-GUSU*), which probably continues a denominal **-nó*-adjective of the type **leuk-s-no-* ‘having light’ (YAv. *raoxšna-* ‘bright’, OHG *liehsen* ‘bright’) based on an **-s*-stem **trég-os* ‘strength’ (cf. Old Norse *þrek n.* ‘id.’). From this **treksno-* a secondary Caland root **treks-* ‘strong’ was derived by suffix deletion, which is at the core of formations such as **treks-iā-* ‘strength’ > Old Irish *treise iā*, f. ‘strength, vigour’, **treks-nt-* ‘id.’ > Mlr. *treisset* ‘strength, might’, the comparative OIr. *tressa*, MW *trech*, MBret. *trech*, NBret. *trec’h* ‘stronger’, and the superlative **treks-(is)mmo-* > OIr. *tressam*, W *trechaf* ‘strongest’ (the British forms have no positive beside them).

An even clearer example is the Celtic neo-root for ‘young’. Like a couple of other branches of Indo-European, Celtic inherited a morphologically complex adjective meaning ‘young’, **h₂j_uh_xǵko-* (cf. Ved. *yuvaśá-*, Lat. *iuuencus*, Proto-Germanic **jungaz*). This formation is not derived from a Caland root, but rather from the noun **h₂óǵi-u-* n. ‘vital force’ (Ved. *áyu-*, OAv. *āiiiū-*). Its Proto-Celtic descendant **ǵi_uanko-* is continued by OIr. *óac*, *óc*, MW *ieuanc*, W *ifanc*, OCorn. *iouenc*, and MBret. *yaouanc* ‘young’.⁹ Based on its status as an essential property concept, this adjective became the basis of its own inner-Celtic Caland system consisting of a comparative and a superlative. In order to generate them, the **-anko-* part of **ǵi_uanko-* (despite not being a Caland suffix) was deleted and a neo-root **ǵi_u-* ‘young’ was extracted serving as the basis for a root-derived comparative OIr. *óa*, MW *ieu*, W *iau*, Bret. *iaou* ‘younger’ and superlative PCelt. **ǵi_u-isamo-* > OIr. *óam*, MW *ieuhaf*, W *ieuaf*, *iefaf*, *ifaf* ‘youngest’.

Taken together, the secure examples of inner-Celtic deradical Caland verbs of group (B) on the one hand, and the nominal neo-roots **treks-* ‘strong’ and **ǵi_u-* on the other seem to corroborate the view that Caland morphology remained mildly productive in Celtic.

8 Conclusion

This study is the first of its kind to provide an exhaustive overview, discussion, and analysis of both inherited and potentially innovative deadjectival verbs in the Celtic languages. The preceding sections reveal that many of these verbs present philological, phonological, and morphological issues, to which one can add the synchronically full or partially weak behaviour of the OIr. verbs *do-lin* (Section 3.4), *sínid* (Section 3.5) and *tóaid* (Section 4.3). In addition, the total number of potential Caland verbs is rather low so that any conclusions drawn from them need to be taken with caution, particularly when attempting to make broad generalisations.

Nevertheless, the investigation has yielded significant findings. Of the twenty Caland-associated verbs, eleven appear to belong to an inherited stock of formations with formal cognates in other Indo-European languages. The remaining nine formations lack exact correspondences in other Indo-European branches, and therefore likely represent innovations within the Celtic languages themselves. Each of these verbs is associated with an adjective (or a substantivisation

⁹Besides PCelt. **ǵi_uanko-*, the Gaulish personal names in *Io(u)inc-* such as *Iouinca*, *Iouincillus*, etc. (cf. Delamarre 2003: 191–192) suggest the existence of a variant **ǵi_uenko-* (i.e., with full grade of the suffix, perhaps in analogy to a stem **ǵi_uen-* < **h₂j_uh_xen-*, inferable from Latin *iuuenis* ‘young (man)’?).

of such an adjective), suggesting that they were derived within Celtic based on these adjectives. Since the formation of these verbal stems follows root-based primary patterns, the suffix of the adjective (e.g., *-to-, *-o-, *-ro-) appears to have been systematically deleted in the formation of these deadjectival verbs.

The deletion of adjectival suffixes and the emergence of primary deradical verbs in these cases further support the hypothesis that Caland morphology maintained a degree of vitality in Celtic. This process of suffix deletion in verbal formations is mirrored in the nominal domain, where adjectives like *treksno- ‘strong’ and *jouanko- ‘young’ similarly give rise to neo-Caland roots *treks- and *jou-, which in turn serve as the basis for further deadjectival derivation.

These patterns, found in both the verbal and nominal domains, suggest the persistence and a certain degree of flexibility of the Caland system within the Celtic languages. Further research could explore whether similar processes of suffix deletion and root creation occurred in other Indo-European branches as well, providing a more comprehensive understanding of the Caland phenomenon across the language family.

Abbreviations

For quotations from Irish and Welsh, we follow the conventions and abbreviations in eDIL (version 2019) and GPC.

Act.	Active	OAv.	Old Avestan
Bret.	Breton	OBret.	Old Breton
Corn.	Cornish	OCorn.	Old Cornish
Dat.	Dative	OIr.	Old Irish
Gaul.	Gaulish	OW	Old Welsh
Gen.	Genitive	PCelt.	Proto-Celtic
Lat.	Latin	Perf.	Perfect
Latv.	Latvian	Pl.	Plural
Lith.	Lithuanian	PrimIr.	Primitive Irish
MBret.	Middle Breton	Ptcpl.	Participle
MCorn.	Middle Cornish	Pres.	Present
MIr.	Middle Irish	Pret.	Preterite
ModW	Modern Welsh	Sg.	Singular
MW	Middle Welsh	Tr.	Transitive
NBret.	Modern Breton	Ved.	Vedic
Nom.	Nominative	W	Welsh

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